



Time Series Analysis

Seminar 1: Stochastic processes and stationarity



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Acronyms used in this material

- ACF: Autocorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BET: Bucharest Exchange Trading (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- EODHD: EOD Historical Data (market-data provider)
- GDP: Gross Domestic Product
- HICP: Harmonised Index of Consumer Prices
- MA: Moving Average
- PACF: Partial AutoCorrelation Function

Today's question and route

- **Question:** is a series stationary, is it white noise, and how do the ACF and the Ljung–Box test answer these questions?
 - ▶ this seminar comes **before** Lecture 1: the section “Prerequisites for today” gives every definition the tasks use
- **Route**
 - ▶ Part A: autocovariances of moving averages and of a random walk, a sample ACF and a Ljung–Box statistic by hand
 - ▶ Part B: BET, S&P 500 and EUR/RON returns; Romanian GDP and inflation, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	mean, autocovariances and ACF of MA(1) and MA(2) processes	Solved, Proposed	A1
A3, A4	a random walk; a trend-stationary series and its first difference	Solved, Proposed	A3
A5, A6	sample ACF, bands, Box–Pierce and Ljung–Box by hand	Solved, Proposed	A5
B1, B2	prices and returns: BET; S&P 500 and EUR/RON	Solved, Proposed	B1
B3, B4	transformations: Romanian real GDP; Romanian HICP inflation	Solved, Proposed	B3
C1, C2	has inflation become more persistent? what is wrong in an AI answer?	Proposed	B1, B4

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
BET, S&P 500	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, not adjusted	1995–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	1996–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped; the last day is 18 September 2026
- Growth and inflation rates in %: $100 \Delta \ln Y_t$ (one period), $100 \Delta_4 \ln Y_t$ and $100 \Delta_{12} \ln P_t$ (one year)
- In the notebook: `b1_returns('bet')`, `sample_acf(x)`, `ljung_box(x, 10)`; no account or key is needed

Prerequisites for today (1/5): processes and moments

- ▣ **Stochastic process** $\{X_t\}$: one random variable for each date t ; a time series $x_1, \dots, x_T$ is one observed path
 - ▶ mean $\mu_t = E[X_t]$; autocovariance $\gamma(t, s) = \text{Cov}(X_t, X_s)$
 - ▶ $E[\cdot]$: the expected value, an average over all possible paths at a fixed date; Cov: the covariance
- ▣ **Weakly stationary**: $E[X_t^2] < \infty$, $E[X_t] = \mu$ and $\text{Cov}(X_t, X_{t+h}) = \gamma(h)$ for every t
 - ▶ h : the lag, the distance in periods between two dates; $\gamma(0) = \text{Var}(X_t)$
 - ▶ **ACF**: $\rho(h) = \gamma(h)/\gamma(0)$, $\rho(0) = 1$, $|\rho(h)| \leq 1$, $\rho(-h) = \rho(h)$
- ▣ Rules for covariances (with constants a, b, c):
 - ▶ $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$; $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$
 - ▶ $\text{Var}(\sum_i a_i X_i) = \sum_i \sum_j a_i a_j \text{Cov}(X_i, X_j)$

Prerequisites for today (2/5): white noise, moving averages, random walk

- **White noise** $\varepsilon_t \sim \text{WN}(0, \sigma^2)$: mean 0, variance σ^2 , $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$
 - ▶ weak (uncorrelated), i.i.d. (independent), Gaussian (i.i.d. Normal)
- **MA(q)** (moving average): $X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$
 - ▶ μ : the mean; $\theta_1, \dots, \theta_q$: the weights of the past shocks; q : the order, the number of past shocks
 - ▶ with $\theta_0 = 1$: $\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h}$ for $0 \leq h \leq q$, and $\gamma(h) = 0$ for $h > q$
- **Random walk**: $X_t = X_{t-1} + \varepsilon_t$, $X_0 = 0$, so $X_t = \varepsilon_1 + \dots + \varepsilon_t$; with drift: $X_t = c + X_{t-1} + \varepsilon_t$
 - ▶ c : the drift, the average change per period
 - ▶ **difference**: $\Delta X_t = X_t - X_{t-1}$; **seasonal difference**: $\Delta_s X_t = X_t - X_{t-s}$, s : the number of periods in a year

Prerequisites for today (3/5): sample ACF and its bands

- ▣ Sample mean $\bar{x}$, **sample autocovariance** and **sample ACF** (divisor T):
 - ▶ $\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (x_t - \bar{x})(x_{t+h} - \bar{x})$, $\hat{\rho}(h) = \hat{\gamma}(h)/\hat{\gamma}(0)$
- ▣ If the series is i.i.d.: $\hat{\rho}(h) \approx N(0, 1/T)$; **95% band** $\pm 1.96/\sqrt{T}$
 - ▶ 1.96: the 97.5% quantile of the standard Normal distribution
 - ▶ a bar outside the band: $\rho(h) \neq 0$ at 5%, for that lag alone; with 20 lags, about 1 bar crosses by chance
- ▣ Typical patterns
 - ▶ white noise: no bar outside; MA(q): bars up to lag q , then nothing
 - ▶ AR(1), $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$: $\rho(h) = \phi^h$, geometric decay
 - ▶ random walk or trend: very slow, almost linear decay; seasonal data: peaks at lags $s, 2s, \dots$

Prerequisites for today (4/5): portmanteau tests

- H_0 : $\rho(1) = \dots = \rho(m) = 0$, the series is white noise up to lag m ; under H_0 both statistics are approximately $\chi^2(m)$
 - ▶ **Box–Pierce**: $Q(m) = T \sum_{h=1}^m \hat{\rho}(h)^2$; **Ljung–Box**: $Q^*(m) = T(T+2) \sum_{h=1}^m \hat{\rho}(h)^2 / (T-h)$
 - ▶ both add up the squared sample autocorrelations of lags 1 to m : large values mean that some autocorrelation is not zero
 - ▶ m : the number of lags tested; the factor $(T+2)/(T-h)$ corrects the small-sample behaviour of Q
- **Decision**: reject H_0 at 5% if the statistic exceeds $\chi_{0.95}^2(m)$, the 95% quantile of the chi-square distribution with m degrees of freedom
 - ▶ 5% critical values: $\chi_{0.95}^2(2) = 5.99$, $\chi_{0.95}^2(3) = 7.81$, $\chi_{0.95}^2(8) = 15.51$, $\chi_{0.95}^2(10) = 18.31$, $\chi_{0.95}^2(12) = 21.03$
 - ▶ equivalently: reject if the p-value, the probability under H_0 of a statistic at least this large, is below 0.05

Prerequisites for today (5/5): transformations

▣ **Transformations** of a positive series Y_t

- ▶ log: $\ln Y_t$ stabilises a variance that grows with the level; $100 \Delta \ln Y_t \approx$ growth in %
- ▶ differences remove trends and random walks; Δ_4 and Δ_{12} remove the season of quarterly and monthly data
- ▶ Box-Cox: $(Y_t^\lambda - 1)/\lambda$, with $\lambda = 0$ meaning $\ln Y_t$; λ : the power that is chosen so that the variance becomes stable

A1: an MA(1) process [Solved]

Task

- ▣ $X_t = \varepsilon_t + 0.5 \varepsilon_{t-1}$, with $\varepsilon_t \sim \text{WN}(0, 2)$.
- ▣ 1. Compute the mean $E[X_t]$.
- ▣ 2. Compute $\gamma(0)$, $\gamma(1)$ and $\gamma(2)$.
- ▣ 3. Compute $\rho(1)$ and $\rho(2)$.
- ▣ 4. Say whether the process is weakly stationary, and why.
- ▣ Report: five numbers and one sentence.

Solution

- ▣ 1. $E[X_t] = 0 + 0.5 \cdot 0 = 0$
- ▣ 2. $\gamma(0) = \text{Var}(\varepsilon_t) + 0.25 \text{Var}(\varepsilon_{t-1}) = 2 + 0.5 = 2.50$
- ▣ $\gamma(1) = \text{Cov}(\varepsilon_t + 0.5\varepsilon_{t-1}, \varepsilon_{t+1} + 0.5\varepsilon_t) = 0.5 \cdot 2 = 1.00$;
 $\gamma(2) = 0$ (no common shock)
- ▣ 3. $\rho(1) = 1.00/2.50 = 0.40$; $\rho(2) = 0$
- ▣ 4. Yes: the mean and all $\gamma(h)$ are finite and do not depend on t .



A2: an MA(2) process [Proposed]

Solution discussed in class

Task

- ▣ $X_t = 2 + \varepsilon_t + 0.4 \varepsilon_{t-1} - 0.3 \varepsilon_{t-2}$, with $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A1.
- ▣ 1. Compute the mean.
- ▣ 2. Compute $\gamma(0)$, $\gamma(1)$, $\gamma(2)$ and $\gamma(3)$.
- ▣ 3. Compute $\rho(1)$ and $\rho(2)$, and sketch the ACF up to lag 4.
- ▣ 4. Say how the constant 2 changes the autocovariances.
- ▣ Report: six numbers, the sketch and one sentence.



A3: a random walk [Solved]

Task

- ▣ $X_t = X_{t-1} + \varepsilon_t$, $X_0 = 0$, $\varepsilon_t \sim \text{WN}(0, 0.25)$.
- ▣ 1. Compute $E[X_{100}]$, $\text{Var}(X_{100})$ and $\text{Var}(X_{120})$.
- ▣ 2. Compute $\text{Cov}(X_{100}, X_{120})$ and $\text{Corr}(X_{100}, X_{120})$.
- ▣ 3. Say whether X_t and ΔX_t are stationary.
- ▣ 4. With a drift $c = 0.1$, compute $E[X_{100}]$.
- ▣ Report: six numbers and two sentences.

Solution

- ▣ 1. $X_t = \sum_{i=1}^t \varepsilon_i$: $E[X_{100}] = 0$; $\text{Var}(X_{100}) = 100 \cdot 0.25 = 25$; $\text{Var}(X_{120}) = 30$
- ▣ 2. The common shocks are $\varepsilon_1, \dots, \varepsilon_{100}$: $\text{Cov} = 25$; $\text{Corr} = 25 / \sqrt{25 \cdot 30} = \sqrt{100/120} = 0.913$
- ▣ 3. X_t : no, its variance grows with t ; $\Delta X_t = \varepsilon_t$: yes, it is white noise.
- ▣ 4. $E[X_{100}] = 100 \cdot 0.1 = 10$; the variance does not change.



A4: a trend-stationary series [Proposed]

Solution discussed in class

Task

- ▣ $Y_t = 5 + 0.2t + \varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A3.
- ▣ 1. Compute $E[Y_t]$ and $\text{Var}(Y_t)$, and say whether Y_t is stationary.
- ▣ 2. Write $D_t = \Delta Y_t$ and compute its mean, $\gamma(0)$, $\gamma(1)$ and $\rho(1)$.
- ▣ 3. Compare Y_t with a random walk with drift 0.2: what happens to a shock ε_{50} in each?
- ▣ Report: five numbers and two sentences.



A5: a sample ACF by hand [Solved]

Task

- ▣ Eight observations: $x = (4, 6, 5, 8, 7, 9, 6, 7)$.
- ▣ 1. Compute $\bar{x}$ and the deviations $x_t - \bar{x}$.
- ▣ 2. Compute $\hat{\gamma}(0)$, $\hat{\gamma}(1)$, $\hat{\gamma}(2)$ with divisor $T = 8$, then $\hat{\rho}(1)$ and $\hat{\rho}(2)$.
- ▣ 3. Compute the band $\pm 1.96/\sqrt{T}$ and compare.
- ▣ 4. Compute the Ljung–Box $Q^*(2)$ and decide at 5%.
- ▣ Report: seven numbers and one sentence.

Solution

- ▣ 1. $\bar{x} = 52/8 = 6.5$; deviations $(-2.5, -0.5, -1.5, 1.5, 0.5, 2.5, -0.5, 0.5)$
- ▣ 2. sum of squares 18: $\hat{\gamma}(0) = 2.250$; products at lag 1 sum to 0.25: $\hat{\gamma}(1) = 0.0312$; at lag 2 to 7.00: $\hat{\gamma}(2) = 0.875$
- ▣ $\hat{\rho}(1) = 0.014$, $\hat{\rho}(2) = 0.389$
- ▣ 3. Band ± 0.693 : both inside
- ▣ 4. $Q^*(2) = 8 \cdot 10 (\hat{\rho}(1)^2/7 + \hat{\rho}(2)^2/6) = 2.02 < 5.99$ ($p = 0.36$): do not reject white noise; with $T = 8$ the test has almost no power.

A6: Box–Pierce or Ljung–Box? [Proposed]

Solution discussed in class

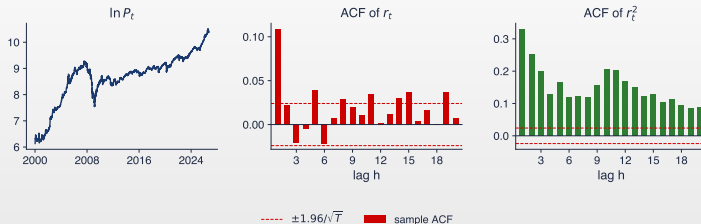
Task

- ▣ A series of $T = 120$ monthly returns has $\hat{\rho}(1) = 0.21$, $\hat{\rho}(2) = -0.08$, $\hat{\rho}(3) = 0.12$. Model: A5.
- ▣ 1. Compute the band and say which autocorrelations lie outside it.
- ▣ 2. Compute $Q(3)$ and $Q^*(3)$.
- ▣ 3. Compare both with $\chi^2_{0.95}(3) = 7.81$ and decide.
- ▣ 4. Explain why the two tests can disagree, and which one you trust here.
- ▣ Report: four numbers and two sentences.

B1: BET prices and returns [Solved]

- **Question:** is the BET log price stationary, and are BET daily returns white noise?
- **Inputs:** BET closes since 2000; $\ln P_t$ and $r_t = 100 \Delta \ln P_t$
- **Tasks**
 1. Compute the sample ACF of $\ln P_t$ at lags 1, 10 and 50.
 2. Draw the ACF of r_t up to lag 20 with the band $\pm 1.96/\sqrt{T}$, and count the bars outside it.
 3. Compute Ljung–Box $Q^*(10)$ for r_t and for r_t^2 , with their p-values.
 4. Interpretation: can yesterday's return be used to forecast today's return?
- **Report:** three ACF values, the chart, two statistics with p-values and two sentences
- **Notebook:** section B1

B1: solution [Solved]



- $\ln P_t$: $\hat{\rho}(1) = 0.999$, $\hat{\rho}(10) = 0.991$, $\hat{\rho}(50) = 0.953$: almost no decay, a random-walk-like level, not stationary
- r_t ($T = 6670$, band ± 0.024): 7 of 20 bars outside, the largest $\hat{\rho}(1) = 0.109$; $Q^*(10) = 108.0$, $p < 0.001$
- r_t^2 : $\hat{\rho}(1) = 0.33$; $Q^*(10) = 2440$, $p < 0.001$: much stronger dependence
- Interpretation: yesterday's return explains only $\hat{\rho}(1)^2 = 1.2\%$ of today's variance, too little for a trading rule after costs; the size of yesterday's move does predict today's risk

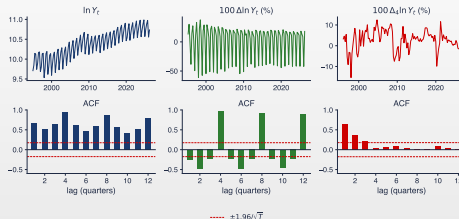
B2: the S&P 500 and EUR/RON [Proposed]

- ▣ **Question:** do the S&P 500 and the EUR/RON rate behave like the BET?
- ▣ **Inputs:** S&P 500 closes since 2000; EUR/RON (BNR reference rate) since July 2005; model: B1
- ▣ **Tasks**
 1. Repeat the three steps of B1 for both series.
 2. Compare the sign and the size of $\hat{\rho}(1)$ of the returns across BET, S&P 500 and EUR/RON.
 3. Compare $Q^*(10)$ of the squared returns across the three series.
 4. Interpretation: which of the three return series is closest to white noise?
- ▣ **Report:** one table (three series, six numbers each) and two sentences
- ▣ **Notebook:** section B2

B3: Romanian real GDP [Solved]

- **Question:** which transformation makes Romanian quarterly GDP closest to stationary?
- **Inputs:** real GDP, chain-linked volumes (2010), not seasonally adjusted, Eurostat, from 1995
- **Tasks**
 1. Compute $\ln Y_t$, $100 \Delta \ln Y_t$ and $100 \Delta_4 \ln Y_t$.
 2. Draw the three series and their ACF up to lag 12.
 3. Report $\hat{\rho}(1)$, $\hat{\rho}(4)$ and $Q^*(8)$ for each.
 4. Interpretation: why is $\hat{\rho}(4)$ of the quarterly growth rate close to 1?
- **Report:** a table of nine numbers, the chart and two sentences
- **Notebook:** section B3

B3: solution [Solved]



	$\hat{\rho}(1)$	$\hat{\rho}(4)$	$Q^*(8)$
$\ln Y_t$	0.68	0.94	495.0
$100 \Delta \ln Y_t$	-0.24	0.96	329.6
$100 \Delta_4 \ln Y_t$	0.63	0.03	74.2

- $\ln Y_t$: trend and season, peaks at lags 4 and 8; $\Delta \ln Y_t$: the season dominates (standard deviation 27.5%); $\Delta_4 \ln Y_t$: mean 2.73%, the ACF dies out after 2–3 quarters
- Interpretation: a q/q rate of unadjusted data repeats the same seasonal swing every year, hence $\hat{\rho}(4) \approx 1$; the annual rate removes the season but reacts late to turning points and overlaps across quarters (positive $\hat{\rho}(1)$)

B4: Romanian inflation [Proposed]

- **Question:** is monthly Romanian inflation since 2005 white noise around its mean?
- **Inputs:** HICP, 2015 = 100, Eurostat, January 2005 to the last month available; model: B3
- **Tasks**
 1. Compute the monthly inflation $100 \Delta \ln P_t$ and the 12-month inflation $100 \Delta_{12} \ln P_t$.
 2. Draw the ACF of both up to lag 36 and report $\hat{\rho}(1)$, $\hat{\rho}(12)$ and $\hat{\rho}(24)$.
 3. Compute Ljung–Box $Q^*(12)$ for the monthly inflation.
 4. Interpretation: what does the ACF of the monthly inflation say about how fast an inflation shock fades?
- **Report:** six ACF values, one statistic with its p-value, the chart and two sentences
- **Notebook:** section B4

C1: has Romanian inflation become more persistent? [Proposed]

- ▣ **Question:** is monthly inflation more persistent after 2020 than in 2005–2019?
- ▣ **Inputs:** monthly HICP inflation of B4, split in January 2020; models: B1, B4
- ▣ **Tasks**
 1. Compute $\hat{\rho}(1)$ and $Q^*(12)$ in each subperiod, with the band of each.
 2. Draw $\hat{\rho}(1)$ on rolling windows of 60 months.
 3. List the events of 2020–2024 that could change the persistence.
 4. Interpretation: why is the difference of two sample autocorrelations hard to judge with 80 observations?
- ▣ **Report:** a table, the chart and a plan for a project
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to interpret the S&P 500 daily data since 2000 ($T = 6\,714$ returns). The answer:
- (a) The ACF of the log price at lag 1 is 0.999, so the log price is a stationary process with a very long memory.
- (b) The return autocorrelation at lag 1 is -0.099, outside the band of 0.024, so a trader can forecast most of tomorrow's return.
- (c) Ljung-Box $Q(10)$ of the squared returns is 6133, so the returns themselves are autocorrelated.
- (d) A weakly stationary process is always strictly stationary, since its mean and variance are constant.
- (e) Differencing a series that is already stationary does no harm: white noise stays white noise.
- (f) A random walk has $\text{Var}(X(t)) = t * \sigma^2$, so it is not stationary.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Autocovariances of linear processes follow from the covariance rules: only common shocks count
- ▣ A random walk has a growing variance; its difference is white noise; a trend-stationary series becomes an over-differenced MA(1)
- ▣ Look at the level, the changes and the squares: prices wander, returns are almost uncorrelated, squares are strongly correlated
- ▣ For macro data, choose the transformation first ($\log$, Δ , Δ_4 , Δ_{12}), then read the ACF
- ▣ An AI answer is a draft: check what is stationary, what “significant” means, and whether squares or levels were tested

After the seminar

- Lecture 1 develops each topic of today: stationarity, white noise and the random walk, Wold, ergodicity, ACF and PACF, the portmanteau tests, transformations
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation persistence in Romania and in the region (Hungary, Poland, Czechia)
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 1–2; [Hyndman and Athanasopoulos \(2021\)](#), Ch. 2–3; [Brockwell and Davis \(2016\)](#), Ch. 1
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

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